

Cookbook TS741

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1 Overview

The cookbook is a simple file that contains all the essential information about the pedal. If you want to recreate it while modifying the sound or changing some parameters, you can refer to this file and easily adjust resistors or capacitors to achieve the desired result.

2 Input Buffer

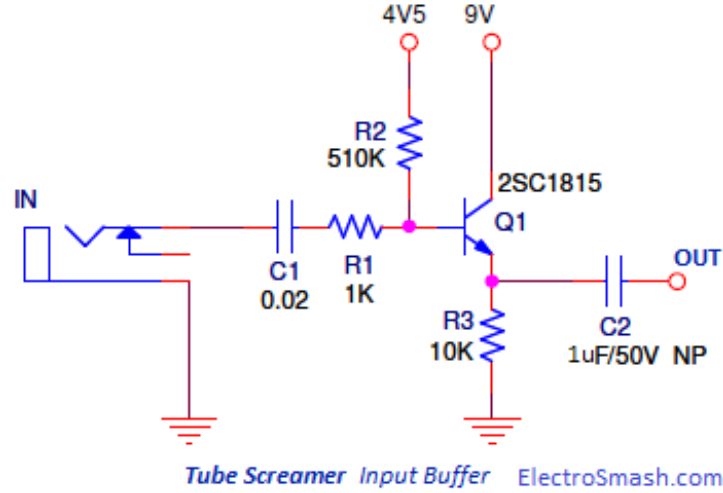


Figure 1: Input buffer

The input buffer preserves the signal as much as possible by providing high input impedance and a unity gain.

2.1 Input Impedance

The input impedance can be calculated using the hybrid- π model for the transistor:

$$Z_{in} = R_1 + (R_2 \parallel [r_{\pi} + (\beta + 1)R_3]) \quad (1)$$

Since the transistor's β parameter is high, the input impedance approaches the value of resistor R_2 . For proper operation, it is recommended to interface the circuit with a guitar having an impedance of at least 500 k Ω .

2.1.1 Voltage Gain

The input buffer gain is approximately unity ($A_v \approx 1$), as its primary function is to preserve the signal without significant amplification.

3 Clipping Stage

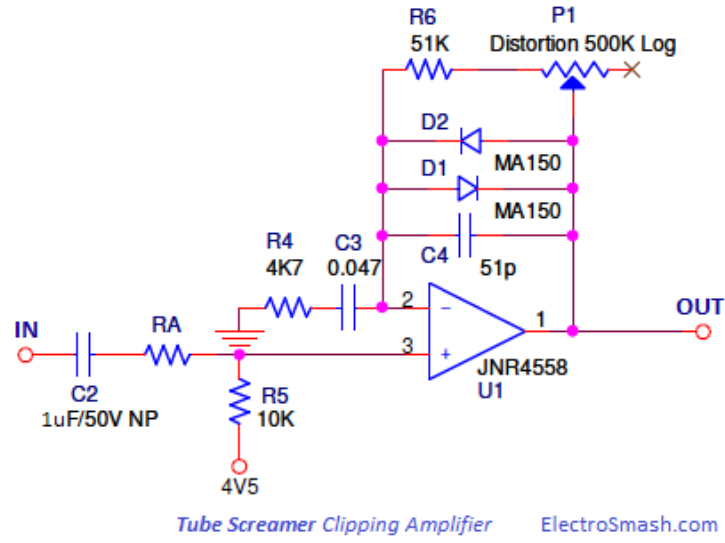


Figure 2: Clipping stage

The clipping stage is responsible for the characteristic distortion of the Tube Screamer. The gain of the operational amplifier in a non-inverting configuration is given by:

$$A_v = 1 + \frac{R_6 + P_1}{R_4} \quad (2)$$

where P_1 is the gain control potentiometer. Since high-value logarithmic potentiometers are not easily available, an effective method to increase the gain is to modify resistor R_6 .

4 Output Buffer

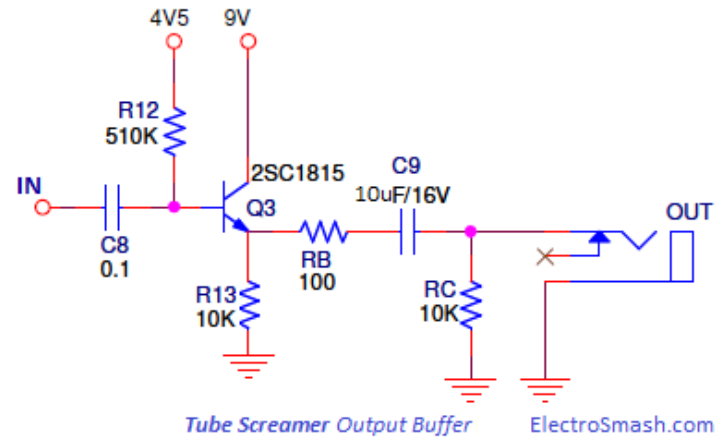


Figure 3: Output buffer

The output buffer reduces the circuit's output impedance, making it suitable for driving subsequent stages without signal degradation.

The output impedance is given by:

$$Z_{\text{out}} = R_C \parallel \left(R_B + \left(R_{13} \parallel \frac{r_{\pi} + R_{12}}{\beta + 1} \right) \right) \quad (3)$$

To ensure a low output impedance, it is advisable to choose a low value for resistor R_C .